THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 11: **Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the Earth's axis were tilted by 45° instead of 23.5°, the seasons would be: 1

(a) the same **(b)** milder **(c)** more extreme **(d)** absent

2. A year in which the Earth makes exactly 366.25 days in an orbit would need: 1

(a) no leap years **(b)** more frequent leap years

(c) same **(d)** fewer leap years

3. A student claims that the Earth's shadow is formed only during lunar eclipses. The correct statement is: 1

(a) it is formed by the Moon

(b) the Earth always has a shadow but the Moon enters it only during eclipses

(c) the shadow forms only then

(d) no shadow exists

4. If the Earth rotated twice as fast, the day would be about: 1

(a) 48 hours **(b)** 24 hours **(c)** 6 hours **(d)** 12 hours

5. The Moon takes 27.3 days to complete a revolution but 29.5 days from full moon to full moon because: 1

(a) the Sun moves **(b)** the Earth moves around the Sun meanwhile

(c) the Moon slows down **(d)** the Moon stops

6. A student claims 'the Sun is closer to Earth in summer'. A good test is: 1

(a) measure the temperature once

(b) look at the Sun

(c) ask friends

(d) compare dates of nearest distance with seasons

7. If the Earth revolved around the Sun in half the time, a year would have about: 1

(a) 365 days **(b)** 182 days **(c)** 91 days **(d)** 730 days

8. The Sun is much larger than the Moon, yet in a total solar eclipse the Moon can hide it because the Moon is: 1

(a) larger **(b)** much hotter

(c) brighter **(d)** much nearer to the Earth

9. A student says that 'the Moon has phases because the Earth's shadow falls on it'. This is: 1

(a) partly correct **(b)** correct
(c) correct for the first quarter **(d)** incorrect

10. If the Earth were upright (no tilt), the Sun at noon on the equator would be: 1

(a) overhead twice **(b)** never overhead
(c) overhead all year **(d)** overhead only in June

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the Moon's light does not produce a clear shadow as strongly as the Sun's. 2

12. Why do the Sun and the Moon rise in the east and set in the west? 2

13. Why do places at high latitudes have very long summer days? 2

14. Why is the Pole star not exactly at the geographic north? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The Earth rotates 360° in 24 h. Find how many degrees it rotates in 1 h and 4 min, and the time for 15° of rotation. 3

16. Explain how a total lunar eclipse might last over an hour but a total solar eclipse only minutes. 3

17. Discuss how the study of eclipses helped scientists, with two examples. 3

OR

If the new moon occurs on 1 March, estimate the dates of first quarter, full moon and last quarter. 3

18. Why does the Sun rise earlier in eastern India than in western India? 3

OR

Explain what happens to day and night at the equator through the year. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Design an experiment using a globe, a lamp and a stick to demonstrate the length of the shadow and the Sun's height at different latitudes. 5

OR

Explain how astronomers predict the time of eclipses using the motions of the Sun, Earth and Moon. 5

20. Explain why total solar eclipses are rare at a given place and plan an observation project of an eclipse, with safety. 5

OR

Describe how the position of sunrise and sunset shifts along the horizon through the year and explain using the tilt. 5

END OF QUESTION PAPER — Total: 40 Marks